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Day 19 — RAG Fundamentals

RAG Fundamentals — Retrieval-Augmented Generation

What is RAG, and why it beats a bare LLM?

Retrieval-Augmented Generation (RAG) is a technique that connects a language model to an external knowledge source at the moment it answers a question, instead of relying only on what it memorized during training. A bare LLM only knows general patterns from its training data: it doesn't know a company's private documents, its latest policies, or anything that changed after training ended. It will often guess confidently and get specifics wrong (a "hallucination"). RAG fixes this by first retrieving the most relevant pieces of real, current information from a company's own documents, then handing that information to the LLM alongside the question, so the model answers based on facts it can actually see rather than facts it is trying to recall from memory.

Ingestion path (building the knowledge base)



Query path (answering a question)



Key ideas

Embeddings	Numeric vectors that capture the meaning of text, so pieces of similar meaning end up close together in vector space.
Similarity search	Finding the stored chunks whose vectors are closest to the question's vector — this is how the "right" context gets retrieved.
Chunking	Splitting long documents into smaller pieces before embedding, so retrieval returns focused, relevant passages instead of whole documents.
Metadata	Extra tags on each chunk (source, date, category) used to filter or prioritize results during search.
Grounding	Anchoring the LLM's answer to retrieved real content, which reduces hallucination and keeps answers verifiable.

Why chunk size matters?

Chunk size is a trade-off. Chunks that are too large (e.g. a whole document) dilute the embedding — the vector represents an average of many ideas, so search becomes imprecise, and stuffing a huge chunk into the prompt wastes context space and can push out other relevant results. Chunks that are too small (e.g. a single sentence) lose surrounding context, so a retrieved fragment may be technically similar but meaningless on its own or missing the qualifier that changes its meaning. Good chunking (commonly a few hundred tokens, sometimes with slight overlap between chunks) keeps each piece focused on one idea while preserving enough surrounding text to be useful once retrieved.

3 real problems in Petalnex-type businesses that RAG would solve

Inconsistent customer support answers — Support agents (or a bot) currently give different answers to the same policy question because knowledge lives scattered across PDFs, spreadsheets, and people's memory. RAG lets every answer pull from the same up-to-date source documents, so responses stay consistent.

Slow product/catalog lookups — Staff waste time searching through catalogs, supplier sheets, or SOPs to answer a simple product, pricing, or availability question. RAG turns that into an instant, natural-language lookup instead of a manual document search.

Onboarding and internal knowledge loss — New employees take a long time to learn internal processes, and knowledge leaves with people who move on. A RAG system built on internal wikis and SOPs gives new hires (and everyone else) a reliable way to ask questions and get answers grounded in the company's actual, current procedures.